Applied Economics

Instrumental Variables

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Winter 2025/26

Motivation

Does having children cause a decrease in working hours?

Single Linear Regression Statistical model for the population:

$$Y_i = \beta_0 + \beta_1 X_{1i} + u_i \quad i = 1, ..., N$$

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library(wooldridge)
data("labsup")
mlr_labsup <- lm(hours ~ kids, data = labsup)
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- Exogeneity $E[u_i|X_{1i}] = 0 \rightarrow \hat{\beta}_1^{OLS}$ is consistent & unbiased.
- Endogeneity $E[u_i|X_{1i}] \neq 0 \rightarrow \hat{\beta}_1^{OLS}$ is inconsistent & biased.

Instrumental Variable (IV) estimation is an approach to get a consistent & unbiased estimator $\hat{\beta}_1$ even if X_{1i} is exogenous $\rightarrow$ causal inference.

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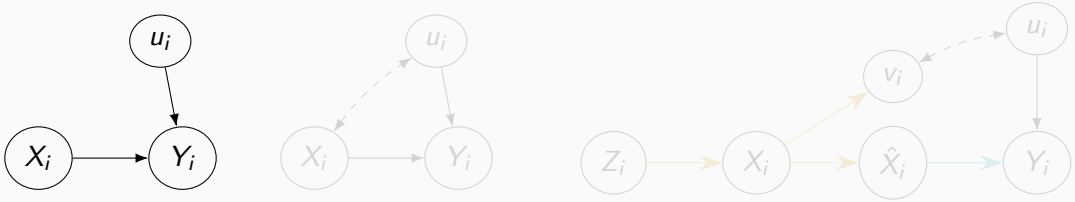
Illustration: Instrumental Variables (IV) approach

X_{1i} is exogenous if it is not correlated with the error term u_i, otherwise endogenous.

Instrumental Variable Z_i

Z_i is **correlated with the endogenous regressor** X_{1i}, **but uncorrelated with** u_i

- First Stage Z_i decomposes X_i's variation into endogenous v_i and exogenous $\hat{X}_i$.
- Second Stage Use X_i's exogenous variation to obtain a consistent estimate $\hat{\beta}^{IV}$.



$\rightarrow$ What correlates with # of kids, but does not correlate with working hours?

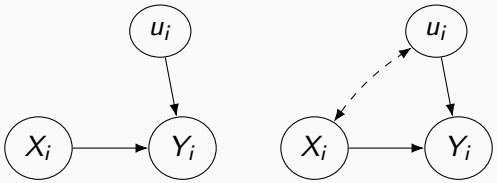
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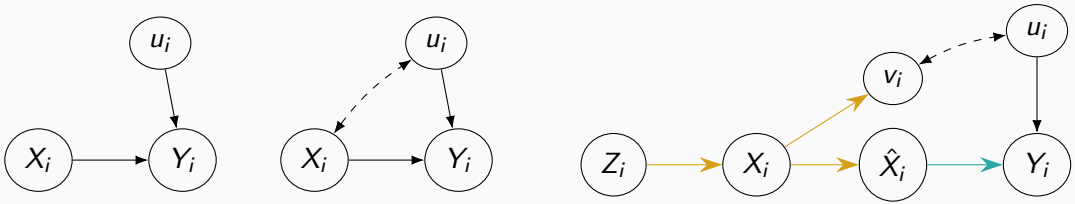
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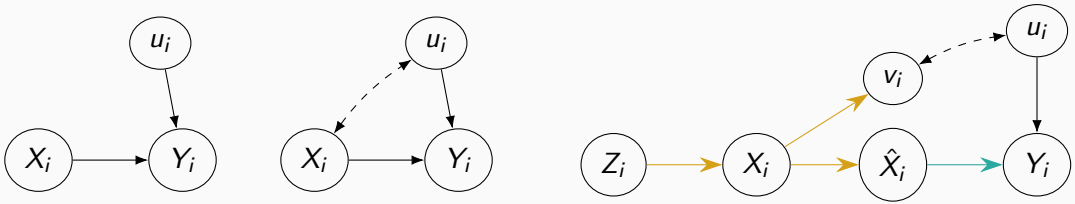
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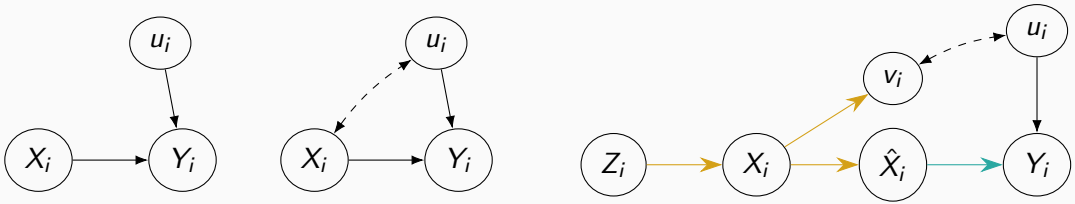
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Regression with one Endogenous Regressor and one Instrument

Instrument Z_i is an indicator variable that turns on if the first two kids are same sex.

$$Y_i = \beta_0 + \beta_1 X_{1i} + u_i \qquad \text{with } E[X_{1i} u_i] \neq 0$$

$$X_{1i} = \pi_0 + \pi_1 Z_i + v_i \qquad \text{with } E[Z_i u_i] = 0$$

A variable Z_i is a **suitable instrument** for X_{1i} if it satisfies two conditions:

1. **Relevance**: $\text{cov}(Z_i, X_{1i}) \neq 0$
   Instrument Z_i "drives" endogenous covariate X_{1i}.
   $\rightarrow$ can be used to extract variation in X_i.
2. **Exogeneity**: $\text{cov}(Z_i, u_i) = 0$
   Instrument is (as if) randomly allocated.
   $\rightarrow$ The variation we extract is **exogenous**.

What does *extract variation* mean?

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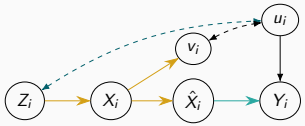
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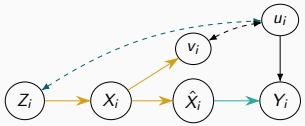
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Two-Stage Least Squares (2SLS)

The Two Stage Least Square estimator **illustrates** the intuition of the IV Approach.

First Stage:

- Test instrument Z_i's **relevance**: $X_{1i} = \pi_0 + \pi_1 Z_i + v_i$.

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first_stage <- lm(kids ~ samesex, labsup)
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- Compute fitted values $\hat{X}_{1i} = \pi_0 + \pi_1 Z_i$.

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labsup$fitted_kids <- first_stage$fitted.values
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Second Stage:

- Estimate regression $Y_i = \beta_0 + \beta_1 \hat{X}_{1i} + u_i$

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2SLS estimator: Properties

Under the assumptions of

- Linearity in parameters (Assumption 1)
- An i.i.d. random sample (Assumption 2)
- Instrument relevance (instead of Assumption 3)
  $\rightarrow$ Testable condition: First Stage.
- Instrument exogeneity (instead of Assumption 4)
  $\rightarrow$ Non-testable condition, needs to be argued.

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IV Identification and Estimation - Derivation

The effect of the instrument Z_i on outcome Y_i is only through regressor X_i:

$$\text{Cov}(Z_i, Y_i) = \text{Cov}(Z_i, \beta_0 + \beta_1 X_i + u_i) = \beta_1 \text{Cov}(Z_i, X_i) + \cancelto{0}{\text{Cov}(Z_i, u_i)}$$

$$\beta_1 = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)}$$

- This identifies the population parameter β_1.
- Using the sample analogue yields the IV Estimator: $\hat{\beta}_1^{IV} = \frac{\widehat{Cov}(Z_i, Y_i)}{\widehat{Cov}(Z_i, X_i)}$

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Variance of the IV Estimator - Intuition

Why do variances of $\hat{\beta}_1^{2SLS}$ and $\hat{\beta}_1^{IV}$ differ?

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coeftest(second_stage, vcov. = vcovHC(second_stage, "HC0"))
summary(slr_iv)
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Variances of $\hat{\beta}_1^{2SLS}$ and $\hat{\beta}_1^{IV}$ under conditional *homoskedasticity* $E[u_i^2 \mid Z_i] = \sigma_u^2$:

$$\text{Var}(\hat{\beta}_1^{2SLS}) = \frac{\sigma_u^2}{n \cdot \sigma_{\hat{X}}^2} \qquad \text{Var}(\hat{\beta}_1^{IV}) = \frac{\sigma_u^2}{n \cdot \sigma_X^2 \cdot \rho_{X,Z}^2}$$

Both in error variance σ_u^2, decrease in sample size n, and in the variance of the regressor σ_X^2. $\hat{\beta}_1^{IV}$ features additional term $\rho_{X,Z}$: the population **correlation** between X and Z.

Since $\rho_{X,Z} < 1$, the IV variance is **larger** $\text{Var}(\hat{\beta}_1^{OLS})$.

- because we only use a **part** of the total variation in X_i in the IV estimation.
- the stronger the correlation between X and Z, the smaller is the increase.

$\rightarrow$ 2SLS procedure underestimates parameter estimate variances!

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Instrument Relevance and Weak Instruments

If the correlation betw. Z_i and X_i is small, we have a weak instrument problem, that yields very imprecise IV estimates

Rule of thumb for sufficiently strong instrument: F-statistic in the first stage > 10.

New research suggests that the threshold value for the F-test should be chosen much larger in practice, ≈ 104.

Weak instruments are also an issue for consistency if exogeneity is not fully satisfied:

$$plim(\hat{\beta}_1^{OLS}) = \beta_1 + Cov(X, u)/Var(X)$$
$$plim(\hat{\beta}_1^{IV}) = \beta_1 + Cov(Z, u)/Cov(Z, X)$$

IV estimator is consistent if exogeneity is satisfied: $Cov(Z, u) = 0$

If exogeneity does not hold, inconsistency of IV is stronger the weaker the correlation of Z and X!

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$$plim(\hat{\beta}_1^{OLS}) = \beta_1 + Cov(X, u)/Var(X)$$
$$plim(\hat{\beta}_1^{IV}) = \beta_1 + Cov(Z, u)/Cov(Z, X)$$

IV estimator is consistent if exogeneity is satisfied: $Cov(Z, u) = 0$

If exogeneity does not hold, inconsistency of IV is stronger the weaker the correlation of Z and X!

Tradeoffs between OLS and IV

If Assumption 4 is violated, we can choose between inconsistent, but precise OLS estimator and a consistent, but less precise IV estimator.

→ **Bias–Variance-Tradeoff** Best choice depends on context.

- Do IV if you think endogeneity is sufficiently important.
  Otherwise, inconsistency of OLS is small, and it might still be close to the true value; whereas IV is correct in infinite samples, but might be pretty far off the true value in finite samples.
- Try to find **strong** instruments, e.g. in the bivariate case an IV where $|corr(X_i, Z_i)|$ is large.

→ In practice, report **both** OLS and IV estimates.

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General Case: Multiple regression model with one or more endogenous regressors

General Case

General linear model with endogeneity: $Y_i = \boldsymbol{X}_i'\boldsymbol{\beta} + u_i$, with $E[\boldsymbol{X}_i u_i] \neq \boldsymbol{0}$

We partition the elements of $\boldsymbol{X}_i$ $(k \times 1)$ into

- endogenous regressors $\boldsymbol{X}_{1i}$ $(k_1 \times 1$ vector) which we try to instrument, and
- exogenous regressors $\boldsymbol{X}_{2i}$ $(k_2 \times 1$ vector)

The exogenous regressors serve as "instruments for themselves":

included instruments $\boldsymbol{Z}_{2i} = \boldsymbol{X}_{2i}$ $(k_2 \times 1)$

For the endogenous regressors, we need variables from outside the model as IVs:

excluded instruments $\boldsymbol{Z}_{1i}$ $(l_1 \times 1)$

We write the $(l \times 1)$-vector of instrumental variables:

$$\boldsymbol{Z}_i = \begin{pmatrix} \boldsymbol{Z}_{1i} \\ \boldsymbol{Z}_{2i} \end{pmatrix}$$

We need at least as many excluded instruments as endogenous regressors: $l_1 \geq k_1$
If $l_1 = k_1$ (as many IVs as endogenous regressors), the model is called *just-identified*.
If $l_1 > k_1$ (more IVs than endogenous regressors), the model is called *over-identified*.

Conditions for suitable IVs

1. Exogeneity: uncorrelated with error u_i
2. Relevance: partially correlated with endogenous explanatory variable

First Stage and Relevance

First stage regression(s): For each endogenous regressor X_{ji}, run a first stage regression of that regressor on all included *and excluded* instruments:

$$X_{ji} = \pi_0 + \underbrace{\pi_1 Z_{1i} + \cdots + \pi_{l_1} Z_{l_1 i}}_{\text{excluded IVs}} + \underbrace{\boldsymbol{X}'_{2i}\gamma}_{\text{included IVs}} + v_i, \qquad j = 1, \ldots, k_1.$$

Estimate the first stage(s)...

```
mlr_first_stage <- lm( kids ~ samesex + age + educ, labsup)
```

... and obtain the fitted values of the endogenous regressor(s): $(\hat{X}_{1i}, ..., \hat{X}_{ki})'$.

```
labsup$fitted_kids <- mlr_first_stage$fitted.values
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Second Stage

In the second stage, we regress the dependent variable Y_i on the predicted values of all endogenous regressors, all exogenous variables and an intercept using OLS.

$$Y_i = \beta_0 + \beta_1 \hat{X}_{1i} + ... + \beta_k \hat{X}_{ki} + \boldsymbol{X}'_{2i}\boldsymbol{\beta}_2 + u_i$$

```
mlr_second_stage <- lm(hours ~ fitted_kids + age + educ, labsup)
coeftest(mlr_second_stage, vcov. = vcovHC(mlr_second_stage, "HC0"))
```

This gives $(\hat{\beta}_0^{2SLS}, \hat{\beta}_1^{2SLS}, ... \hat{\beta}_k^{2SLS})'$, the 2SLS estimates of the model coefficients.

$\rightarrow$ Remember: the previous procedure yields:

- Consistent parameter estimates.
- Incorrect parameter variance estimates.

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IV Estimator

→ Important! use the R package for robust IV for correct variance estimates:

```
mlr_iv <- iv_robust(hours ~ kids + age + educ | samesex + age + educ,
                    data = labsup, se_type = "HC0", diagnostics = TRUE)
summary(mlr_iv)
```